

As for independence, Mandelbrot made a case that even if prices are not autocorrelated, their volatility is correlated over time. This means that big price swings tend to cluster, and it is likely that stocks will move by an above-average amount, even if we do not know the direction of the move. According to Mandelbrot, because the two main assumptions of modern finance are flawed, related models such as CAPM are also flawed. They understate market risk and the amount of capital financial institutions should hold to withstand market risk.

Unfortunately, many academics stopped paying attention to Mandelbrot's financial concepts once they realized that these concepts challenged the usual assumption suspects, which had become part and parcel of most financial models. Mandelbrot's unbounded, nonfinite variance calculations were also difficult to work with.⁹

Despite their many empirical and theoretical challenges, linear factor models do give some indication of the relationship between risk and expected return. Researchers still routinely use linear three- and four-factor models to measure the performance and statistical significance of investment strategies. Factor models may have some value as a general guide to the strength and significance of a strategy. Therefore, we will use them in [Chapter 8](#) as one way to confirm the results of our momentum model.

Warren Buffett, who rivals Yogi Berra as originator of some of the world's best quotes, once said, "Beware of geeks bearing formulas." In terms of developing models that can accurately explain and deal with the real world, financial economists have not been very successful. Researchers have tried to come up with a number of fixes along the way, such as resampling adjustments for MVO and a complicated array of additional factors for CAPM. Effective results are still questionable, at best. In the words of Robert Haugen (2010):

We can advance by developing radically new theories to help us understand what we now see in the data. Or we can go back, denying what is now readily apparent to most, bending the data through ever more convoluted processes until it screams its compliance with our preconceptions.

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